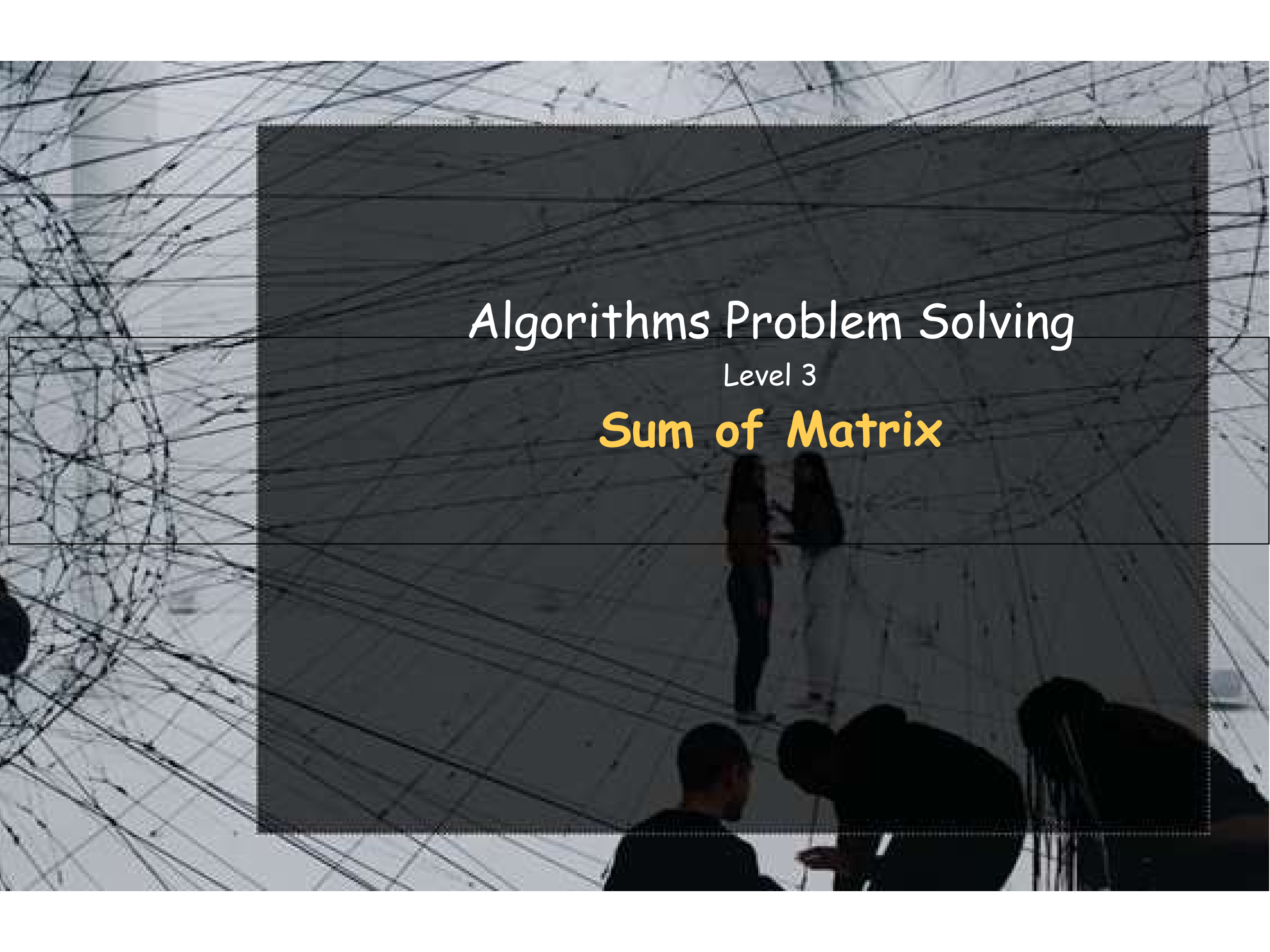


Algorithms Problem Solving

Level 3

Sum of Matrix



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Definition

Matrix Summation

 Program populates a 3×3 matrix with random numbers and calculates the sum of all elements.

Explanation

 Global Accumulation

 Iterates over every row and column, adding `matrix[i][j]` to a total sum variable.

Procedures

1. Create 3×3 matrix and fill with random numbers (1 to 10).
2. Display matrix.
3. Compute total sum using SumMatrix().
4. Print sum result in main().

💡 Example

Matrix1:

04 08 06

03 10 07

08 08 10

➡ Result

Sum of Matrix: 64

Note

- ◆ Ensure accumulator variable is initialized to zero.

⊘ Warning

- ◆ The accumulator variable `sum` must be initialized to `0` before looping.
- ◆ Summing very large matrices or large numbers can exceed `int` data type limits.
- ◆ Matrix dimensions must match the loop bounds to prevent accessing out-of-bounds memory.



Comparison

▨ `SumMatrix()`

- ✓ Accumulates every single element across all rows and columns.
- ✓ Uses double nested loops to visit all $N \times M$ cells.
- ✓ Returns a single global scalar sum value.

▤ `PrintRowSums()`

- ✓ Resets accumulator per row and outputs separate sums.
- ✓ Produces N row totals rather than a single total.
- ✓ Displays intermediate row aggregates.

✦✦ Summary

- ✓ Scans all elements inside a 2D matrix through standard double nested loops.
- ✓ Accumulates every element's numerical value into a single running total variable.
- ✓ Reduces 2D matrix data into a single scalar value representing the total sum.

Thank you 😊